BUSHRA BUSINESS BANK

NETWORK ANALYSIS AND DESIGN

Proposed Enterprise Network Design Report

GROUP MEMBERS

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Design scope: Planning and documentation only — no configuration

1. Executive Summary

This report proposes a secure, scalable and manageable network for Bushra Business Bank. The design uses a hierarchical star topology with a redundant core, separated department VLANs, a protected server segment, an isolated DMZ, secure site-to-site VPN connectivity for two assumed branches, and redundant Internet connectivity. The design is conceptual and intended for academic evaluation; it does not include device configuration, command-line instructions or implementation.

2. Project Assumptions and Scope

Because the bank did not provide an existing network inventory, the group uses the following clearly stated planning assumptions: one head office in Mogadishu and two smaller branches; approximately 150–200 head-office endpoints at initial deployment; approximately 30–50 endpoints per branch; business-critical banking applications hosted in a protected server/data-centre segment; and a future growth allowance. Replace these assumptions if the lecturer or company brief supplies actual figures.

In scope: requirements analysis, logical and physical topology proposal, device roles, VLAN and IP addressing plan, security architecture, resilience, and explanation. Out of scope: router/switch/firewall configuration, CLI commands, cable installation, testing on live equipment, and production deployment.

3. Company Overview

Bushra Business Bank is treated in this academic scenario as a financial institution with a head office and two branches. Its network must support staff communication, customer service, finance operations, secure access to banking applications, branch connectivity, printing, monitoring and controlled guest Internet access.

4. Network Problem Statement

- Department traffic may be mixed on a flat network, increasing unnecessary broadcast traffic and making access control difficult.
- Sensitive Finance, HR and banking systems require stronger separation from general staff and guest devices.
- Branch users need dependable access to central services without exposing internal systems to the public Internet.
- A single network device or Internet connection could create an avoidable service outage.
- Without central monitoring, backups and documented IP allocation, troubleshooting and growth become harder.

5. Proposed Solution and Design Goals

- Use a hierarchical design: edge, core, access and endpoint layers.
- Use department-based VLANs and subnet boundaries to separate traffic.
- Place public-facing services in a DMZ, separate from internal banking systems.
- Use next-generation firewalls, access-control policies, endpoint protection and encrypted VPNs.
- Provide redundant core switches and dual ISP links where budget permits.
- Centralize identity, DNS, DHCP, logging, monitoring and backup services.
- Design with reserved address space to support future departments and branches.

6. Network Topology

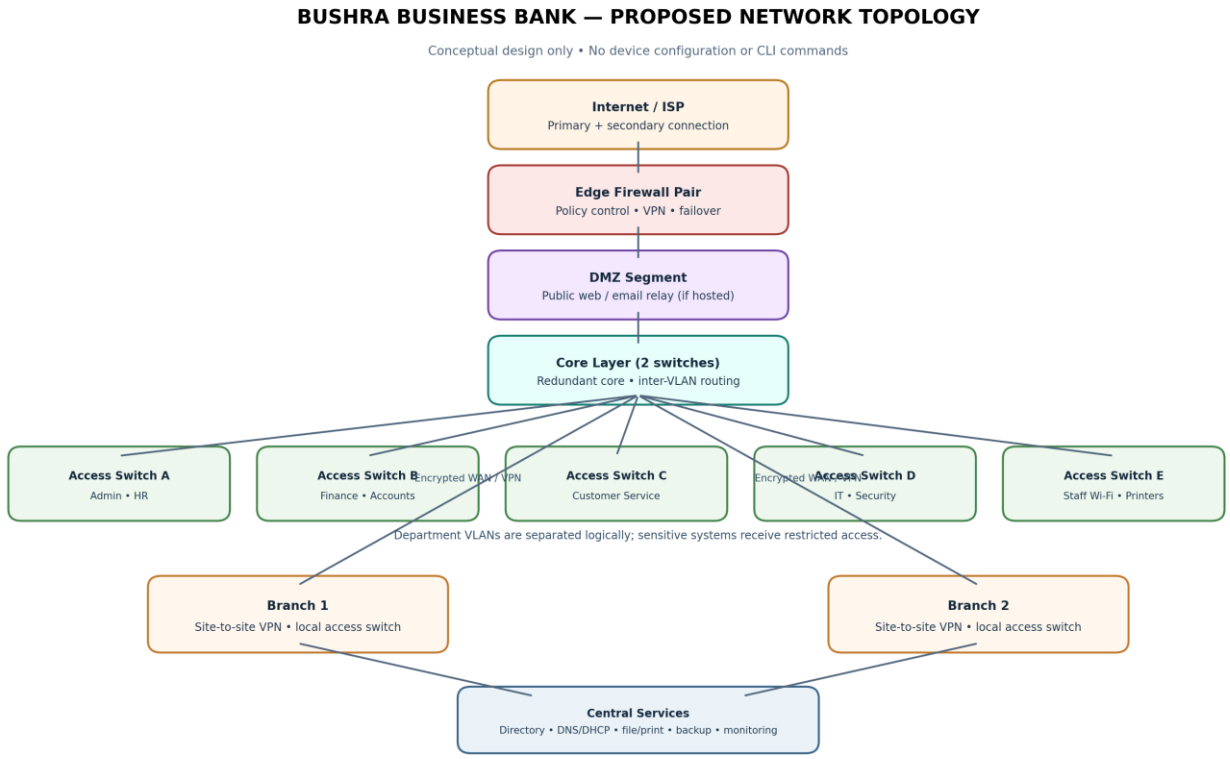


Figure 1. Conceptual topology for the head office and two branches.

Traffic path: Internet service providers connect to the edge firewall pair. The DMZ hosts only approved public-facing services. The core layer connects to department access switches and central services. Branches connect through encrypted site-to-site VPN tunnels. Department VLANs are routed at the core and protected by access-control and firewall policies.

7. Network Devices and Their Roles

Device	Purpose
2 edge firewalls (HA pair)	Perimeter security, VPN termination, policy enforcement and failover.
2 core Layer-3 switches	High-speed backbone, inter-VLAN routing and resilient core connectivity.
Managed access switches	Connect PCs, printers, access points, cameras and phones; support VLANs.
Wireless access points	Provide managed staff Wi-Fi and isolated guest Wi-Fi.
2 ISP connections	Primary and backup Internet paths to improve availability.
Servers / virtual host	Directory and identity, DNS, DHCP, file services, application services, logging and backup.
UPS and rack equipment	Power protection and organized physical installation.
Monitoring / security tools	Central log collection, network monitoring and alerting.

8. VLAN and IP Addressing Plan

Private IPv4 address space 10.10.0.0/16 is reserved for the bank. The table below is a proposed logical plan. The first usable address in each subnet is reserved as the default gateway. Usable host counts exclude network and broadcast addresses. Actual allocation must be checked against the final number of devices before implementation.

VLAN / Site	Department / Purpose	Network / Prefix	Gateway	Usable hosts	Notes
VLAN 10	Administration	10.10.10.0/26	10.10.10.1	62	Staff PCs and admin

					printers
VLAN 20	Finance & Accounts	10.10.20.0/26	10.10.20.1	62	Restricted financial workstations
VLAN 30	Customer Service	10.10.30.0/25	10.10.30.1	126	Teller/customer-service endpoints
VLAN 40	IT Department	10.10.40.0/27	10.10.40.1	30	IT support and management
VLAN 50	Human Resources	10.10.50.0/27	10.10.50.1	30	HR workstations
VLAN 60	Security & CCTV	10.10.60.0/26	10.10.60.1	62	Cameras and physical-security systems
VLAN 70	Staff Wi-Fi	10.10.70.0/25	10.10.70.1	126	Managed staff wireless devices
VLAN 80	Guest Wi-Fi	10.10.80.0/24	10.10.80.1	254	Internet-only guest access
VLAN 90	Servers / Data Centre	10.10.90.0/27	10.10.90.1	30	Directory, DNS, DHCP, file and app servers
VLAN 100	Printers / IoT	10.10.100.0/27	10.10.100.1	30	Network printers and approved IoT
VLAN 110	DMZ	10.10.110.0/28	10.10.110.1	14	Public-facing services, isolated from LAN
VLAN 120	Network Management	10.10.120.0/27	10.10.120.1	30	Management interfaces; IT-only
Branch 1	Branch 1 users	10.10.130.0/24	10.10.130.1	254	Branch user network; VLAN split locally if required
Branch 2	Branch 2 users	10.10.140.0/24	10.10.140.1	254	Branch user network; VLAN split locally if required
WAN links	Point-to-point links	10.10.250.0/30 blocks	Per-link	2 usable each	Allocate a separate /30 per routed point-to-point link

9. Subnetting Explanation

A /26 subnet has 64 total IPv4 addresses and 62 usable host addresses. A /25 has 128 total and 126 usable. A /27 has 32 total and 30 usable. A /28 has 16 total and 14 usable. A /24 has 256 total and 254 usable. Larger subnets are assigned to groups expected to have more devices, such as Customer Service and Wi-Fi. Smaller subnets are assigned to smaller or more sensitive teams, such as IT and HR.

Example: VLAN 20 Finance uses 10.10.20.0/26. Its network address is 10.10.20.0, proposed gateway is 10.10.20.1, usable host range is 10.10.20.1–10.10.20.62, and broadcast address is 10.10.20.63. For endpoint assignment, reserve .1 for the gateway and use a documented DHCP range or static reservations for approved systems.

10. Security Design

- Separate Finance, HR, IT, CCTV, guest wireless, servers and management traffic into distinct VLANs.
- Apply least-privilege access policies: users access only the services required for their role.
- Allow guest Wi-Fi to reach the Internet only; block access to internal bank networks.
- Use MFA for remote and privileged access, strong authentication and role-based permissions.
- Use encrypted site-to-site VPNs between branches and head office.
- Place public-facing systems in a DMZ and restrict DMZ-to-LAN communication.
- Use endpoint protection, timely patching, secure backups and centralized security logging.
- Restrict network-device management to authorized IT workstations and management VLAN.

11. Availability, Performance and Scalability

The proposed design reduces single points of failure through redundant core switches, a high-availability firewall pair and primary/secondary ISP links. Managed switches and structured cabling support orderly expansion. Quality-of-Service policies may prioritize approved real-time business applications. UPS protection and tested backups support continuity. Each branch can be expanded with additional access switches and VLANs while retaining the reserved 10.10.0.0/16 address plan.

12. Implementation Considerations (Design Only)

Before deployment, the bank should confirm the actual number of users, applications, branches, bandwidth needs, existing equipment, regulatory obligations and budget. A detailed site survey and security review should validate the design. The network should be implemented and tested by qualified personnel under an approved change plan. This student project stops at the design stage and intentionally contains no configuration.

13. Conclusion

The proposed Bushra Business Bank network provides a structured foundation for secure communication between departments and branches. VLAN separation, subnet planning, perimeter protection, a DMZ, encrypted VPNs and redundancy address the main assumed network problems while allowing future growth. The design meets the assignment requirement by presenting the topology, IP plan and explanation without performing device configuration.

14. References

- Cisco Networking Academy. Networking Basics and network design learning materials.
- RFC 1918 — Address Allocation for Private Internets.
- RFC 4632 — Classless Inter-domain Routing (CIDR): The Internet Address Assignment and Aggregation Plan.
- Course notes: Network Analysis and Design.